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Dengue Severity Level Prediction Using Hematology-Driven Machine Learning Models

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About Article

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ABSTRACT

Dengue fever represents a significant public health challenge due to its potential to progress rapidly into severe and life-threatening conditions, emphasizing the need for early and reliable severity assessment. Using typically available complete platelet count numbers, this study proposes a hematology-driven machine learning architecture for triage-level dengue severity prediction. An informed severity variable based on clinical considerations was developed to classify the patients into minor, moderate, and severe according to platelet count, hematocrit, white blood cell counts, and hemoglobin based on a freely provided dataset of laboratory confirmed dengue cases. The dataset was obtained from a publicly available Kaggle repository and consists of laboratory-confirmed dengue patient records with complete hematological profiles, totaling 1,450 patient samples. Multi-class severity classification in the case of multi-class classification of severity, four ML algorithm, logistic regression, random forests, support vector machines, and extreme gradient boosting algorithms, trained machine learning models have been trained and evaluated. The extreme gradient boosting model achieved a precision of 64.1% and a macro-averaged F1-score of 0.59, outperforming logistic regression, random forests, and support vector machines by providing more balanced and consistent classification performance across minor, moderate, and severe dengue cases. Although the sensitivity maintained was clinically acceptable, in terms of class-wise analysis, there was a significant predictive ability of the minor and moderate severities. The most salient predictors of the severity were found to be white blood cell count, hematocrit, and platelet count in model interpretability analysis based on Shapley Additive Explanations. These results indicate that machine learning in combination with traditional hematological indicators will offer a viable, cost-efficient, and explainable method of early detecting dengue severity, thus offering meaningful clinical decision support in the medical care environment with scarce resources.

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1. INTRODUCTION

Dengue fever is a rapidly expanding global public health concern, particularly in tropical and subtropical regions where recurrent outbreaks place substantial pressure on healthcare systems. Dengue infection, primarily transmitted by *Aedes* mosquitoes, presents a wide clinical spectrum ranging from mild febrile illness to severe dengue characterized by plasma leakage, hemorrhage, and shock, which can be fatal if not promptly managed (World Health Organization, 2023; World Health Organization, 2021). The unpredictable nature of the dengue development requires timely severity estimation to enable effective triage, maximum resource utilization, and effective patient care. The prediction from WHO shows that millions of dengue cases will be recorded annually, and a considerable number of cases will grow to severe illness because of the early diagnosis or good clinical controlling (World Health Organization, 2023).

As the studies show, such as Khan *et al.* (2023) and Huang *et al.* (2020) have demonstrated ML Models for dengue disease prediction, most of these studies focus on binary prediction or incorporate non-routine clinical, serological, or imaging features, which limit their real-world usage in countries with limited health care resources. Additionally, many existing Models lack explicit clinical interpretability, reducing their usability for triage-level decision-making in real-world healthcare environments.

1.1. Motivation and problem statement

Early detection and estimation of dengue severity is a significant clinical dilemma, especially in low- and middle-income countries where advanced diagnostic modalities and serological tests might be unavailable or simply costly. In accordance with de Oliveira *et al.* (2009), Ralapanawa *et al.* (2018), and Srichaikul & Nimmannitya (2000), conventional clinical assessment addresses might overlook modest hematological changes that indicate fatal sickness outcomes as they primarily rely on physician understanding. Although the broad availability of standard complete blood count indicators such as platelet count, hematocrit, hemoglobin, and white blood cell count, their combined predictive value is often neglected in real-time clinical decision-making. Recent machine learning-based research studies on the topic of dengue severity in resource-limited medical environments have not been widely applied, most of these studies rely on non-routine serological measurements or are more focused on a binary classification (Huang *et al.*, 2020; Chowdhury *et al.*, 2022; Sarma *et al.*, 2022). The importance of an ensemble technique forecasting dengue severity at an early stage is highlighted by these limitations, which can be easily converted into information that is both accurate and interpretable, based on routinely available data.

Unlike prior studies that emphasize laboratory-intensive biomarkers or retrospective severity confirmation, this study targets early-stage, triage-level severity estimation using only routinely available hematological parameters, thereby addressing a critical translational gap between predictive modeling and clinical deployment.

1.2. Objectives of the study

The primary objective of this research is to develop a hematology-driven machine learning model capable of predicting dengue severity levels categorized as minor, moderate, and severe. The focus of the study is to figure out the best model for triage-level severity classification through assessing several supervised learning algorithms applying conventional blood test parameters. One sub-goal is to improve clinical interpretability by using explainable artificial intelligence models, hence clarifying the role of particular hematological features in model forecasts.

1.3. Contributions of the work

This research paper provides a list of critical aspects. First, it introduces a tri-level dengue severity classification method utilizing publicly accessible data and clinically informed hematological constraints. Secondly, it delivers the comparison of state-of-the-art machine learning models for multi-class dengue severity prediction based solely on standard blood parameters. Third, it improves clinical decision-making by providing concise, trustworthy guidance through the integration of explainable machine learning and model interpretability. It uses explainable machine learning to enhance model interpretability, facilitating a transparent and reliable clinical decision-making process.

This proposed study is novel as it provides a single framework where a combination of clinically based tri-level severity labeling, the use of hematological indicators readily available clinically, the use of strong multi-class machine learning comparison, and explainable AI-based interpretability in a resource-limited healthcare environment is used to provide a practical and scalable solution in early dengue severity assessment.

2. LITERATURE REVIEW

Dengue fever, which is spreading geographically and even developing into severe, potentially lethal forms, has long been considered a serious threat to global health. To reduce mortality and the burden on healthcare systems, the World Health Organization continues to emphasize the rising cases of dengue worldwide and the urgent need to identify it early and estimate its severity (World Health Organization, 2023; World Health Organization, 2021). The clinical management of dengue is problematic due to the unpredictability of the disease course and the possibility of rapid worsening despite initially moderate symptoms.

The major objective of the initial clinical studies included identifying hematologic pathologies of dengue infection. In order to be able to differentiate dengue and dengue hemorrhagic fever, Srichaikul and Nimmannitya (2000) emphasized the diagnostic value of hematologic indicators such as thrombocytopenia, hemoconcentration and leukopenia. These findings were supported by findings of the next studies that showed that differences in hematocrit and platelet count are important indicators of the severity of the disease (de Oliveira *et al.*, 2009). Despite these research studies demonstrating that hematologic parameters were clinically relevant to markers,



their predictive ability was limited as most of them depended mainly on descriptive or simple statistical analyses.

The diagnostic utility of routine hematologic clues in the diagnosis of dengue has been reviewed in later observational studies. Both leukopenia and concomitant increases in the hematocrit are useful biomarkers of the initial dengue cases, as reported by Chaloeuwong and Tantiworawit (2018); however, their study did not include the automated severity categorization and triage-level prediction. The increased significance of complete blood count parameters to predict the severity of dengue was also highlighted by Ralapanawa *et al.* (2018); nevertheless, they found that the use of manual interpretation is not even and mostly relies on clinical judgment.

Due to the advancement in artificial intelligence, machine learning algorithms have been widely utilized in prediction and forecasting of dengue. Sarma *et al.* (2020) used machine learning methods to identify dengue-positive and dengue-negative cases and achieved promising results in regard to accuracy; however, their study lacked the stratification of severity since it was restricted to binary classification. Trying to evaluate the risks of severe dengue, Huang *et al.* (2020) combined laboratory data with demographic information in a machine-learning model that overcame traditional statistics. However, their reliance on serological markers limits the applicability to resource limited environments, where these assays are often unavailable.

Efforts to predict dengue severity using machine learning have increased across different demographics and geographical regions. Although the model has poor generalizability to adult populations and limited predictive accuracy, Phakhounthong *et al.* (2018) applied classification and regression tree models to pediatric dengue cases and obtained interpretable results. An ensemble model based on XG boosting is used to predict the severe dengue in Puerto Rico, Madewell *et al.* (2025) reported excellent discriminative performance. Nevertheless, their examination's dependence on complicated clinical features and region-specific data contained wider clinical use.

The importance of the explainability and interpretability of dengue prediction models has been recognised in recent research. Chowdhury *et al.* (2022), using a machine learning framework based on Shapley Additive Explanations, reported that three parameters can be identified as most salient for predicting dengue severity: hematocrit and platelet count. Although this study did increase the level of algorithmic transparency, its reliance on serological test variables may limit its application in resource-constrained healthcare settings. Khan *et al.* (2023) in his study used traditional hematological-based feature and further shows the effectiveness of ML models to predict disease prediction.

In recent reviews, emphasis has been put on the pros and cons of machine learning implementation for dengue prediction at a larger scale. Many extant models have been characterised by the lack of generalisability, interpretability and real-time clinical applicability in their clinical applicability as evidenced by Sarma *et al.* (2022). Hoyos *et al.* (2025) found, in a recent publication, other potential challenges of unbalanced datasets and the dominance of binary classification frameworks in dengue prognostic research. Furthermore, the use of hybrid and deep learning methodologies for epidemic prediction and

severity determination is still scarce in daily clinical practice, considering their frequent use of data from environmental, demographic or non-routine clinical information (Mahanta *et al.*, 2024).

Large-scale surveillance has always been the focus of public health-oriented instead of a clinical decision support at an individual level. To predict the severity of dengue in hyperendemic regions, Salazar-Flórez *et al.* (2025) and developed machine learning models. These models were useful deriving epidemiological inputs, but had a no triage level resolution for bedside decision making. A dengue infection severity score with limited automation and mediocre predictive accuracy was put forth by Pongpan *et al.* (2013) using logistic regression.

Here continue to be plenty of gaps in the body of existing literature despite significant developments. Many studies lack interpretability mechanisms, focus on binary severity outcomes, utilize serological or non-routine results from laboratories, or overlook to address practical deployment in regions with scarce resources. Real-world implementation continues to be constrained by the lack of multi-level, explainable, hematology-based machine learning frameworks suitable for clinical decision assistance.

By developing a hematology-driven, multi-class dengue severity prediction model using only routine complete blood count records, the current work addresses in these gaps in the literature. This work proposes to develop an interpretable, cost-effective, and clinically applicable decision support framework which is compatible with the expectations of resource-constrained healthcare settings by incorporating explainable artificial intelligence approaches and concentrating on triage-level classification.

3. METHODOLOGY

This study develops and assesses a hematology-driven model for early dengue severity level prediction using supervised machine learning algorithms. The overall workflow consists of dataset acquisition, severity labeling of each record, data preprocessing, model development, performance evaluation, and model interpretation. The methodology has been developed focusing on clinical relevance to ensure scientific rigor and reproducibility using parameters that are commonly available through blood laboratory analysis.

The dataset for this study was taken from an easily accessible Kaggle dataset of dengue cases with confirmed laboratory results (Hasan Moon, 2023). Ethical approval was not needed due to the anonymous nature of the data, also including basic complete blood count parameters, such as platelet number, hematocrit proportion, hemoglobin concentration, white blood cell number, red blood cell number, and mean corpuscular volume. In order to provide support for supervised learning and facilitate risk stratification covering towards triage the Severity Level target variable was constructed. This variable supports the classification of cases as minor, moderate or severe according to clinically accepted threshold values for key hematological indicators since explicit severity labels were not available in the original dataset.

The severity labeling process was guided by clinically established hematological thresholds reported in prior dengue



literature and WHO-aligned clinical practice, enabling a reproducible and medically informed tri-level categorization scheme. The thresholds used for assigning minor, moderate, and severe classes are summarized in Table 1.

Table 1. Hematological threshold criteria used for dengue severity labeling

Severity Class	Platelet Count ($\times 10^3/\mu\text{L}$)	Hematocrit (%)	White Blood Cell Count ($\times 10^3/\mu\text{L}$)	Hemoglobin (g/dL)
Minor	≥ 100	< 40	4 – 11	Normal range
Moderate	50 – 99	40 – 45	< 4 or > 11	Mild deviation
Severe	< 50	> 45	Marked leukopenia or leukocytosis	Significant deviation

These threshold values were applied in combination to construct the Severity Level variable, ensuring alignment with clinical indicators of plasma leakage, bleeding risk, and disease progression.

The dataset went through the preprocessing in order to make it suitable for machine learning analysis before training the model. This preprocessing occurred by encoding severities for supervised classification, standardizing hematological variables for continuous response that eliminated bias due to the scale of a measure, and handling and assessing missing values. In order to maintain balanced class distributions, the dataset obtained after the cleaning process was then split into training and testing subsets using a stratified split. The missing data were managed by removing the records that contained incomplete critical hematological parameters because they would have compromised clinical validity and create an artificial imputation bias. To standardize the continuous variables through feature scaling, z-score standardization was employed to normalize the variables and provide equal contribution between features in training the model, especially the distance-based and margin-based classifiers. For the prediction of multi-class dengue severity, four supervised machine learning algorithms namely Logistic Regression, Random Forest, Support Vector Machine and Extreme Gradient Boosting were implemented and trained using training set respectively. Standard metrics for classification, such as accuracy, precision, recall and F1--score were used to assess model performance on the unseen test data. A comparative analysis of the models found the most reliable, most consistent approach to triage -level dengue severity classification.

Explainable AI approaches were leveraged with the best performing model to increase clinical usability as well as trust. To determine the effect of certain hematological features on the predictions of severity a feature contribution analysis was undertaken. This interpretability phase promotes the transparent decision programming in a clinical decision support system and ensures that the model outputs are consistent with the clinical knowledge.

Finally, the methodology is based on a directed pipeline which involves gathering hematological data, clinical severity labeling, preprocessing, supervised model learning, comparative valuation, and the explainable interpretation of the model. This approach covers all possible aspects and ensures that the model chosen for pre-clinical assessment of dengue severity using traditional laboratory data is suitable for practice, relevant to the disease and technically feasible.

3.1. Mathematical equations and symbols

This research using four of the ML algorithms to forecast dengue severity across multiple subclasses, using hematological features. Below are the models' mathematical formulations and the evaluation metrics used.

3.1.1. Logistic regression

Logistic Regression uses a sigmoid function to determine the probability whether a sample will fall into a specific severity class:

$$P(y = k | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_{1x_1} + \beta_{2x_2} + \dots + \beta_{n x_n})}}$$

Where,

$X = (x_1, x_2, \dots, x_n)$ represents haematological features, β_i are model coefficients, and y denotes the predicted severity class.

3.1.2. Random forest

The ensemble learning methodology Random Forest gathers the predictions that are done by different decision trees. The current classification is decided through such a majority voting mechanism:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\}$$

Where,

$h_t(x)$ is the prediction of the t -th decision tree and T is the total number of trees.

3.1.3. Support vector machine

Support Vector Machine constructs an optimal separating hyperplane by maximizing the margin between classes. Function for decision making in this algorithm can be written as:

$$f(x) = \text{sign}(w \cdot x + b)$$

where w is the weight vector, x represents input haematological features, and b is the bias term. Kernel functions are used to model non-linear class boundaries.

3.1.4. Extreme gradient boosting (XGBoost)

XGBoost develop an ensemble of decision trees sequentially, minimizing a regularized objective function:

$$\text{obj} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

Where,

$l(y_i, \hat{y}_i)$ is the loss function, f_k represents the k -th decision tree, and Ω is a regularization term that controls model complexity and reduces overfitting.



3.1.5. Evaluation metrics for the model

Multi-class classification metrics were used to assess the model's performance:

i. *Accuracy*:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

ii. *Precision*:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

iii. *Recall*:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

iv. *F1-score*:

$$\text{F1 - score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

4. RESULTS AND DISCUSSION

This section critically assesses the efficacy of the machine learning approach based on hematology in predicting levels of dengue severity. The key goals are a comparative evaluation of the model predictive ability, ability to discriminate different severity classes at the level of triage, and clinical interpretability of the results. Emphasis is laid on the usefulness of the findings in real life, especially in terms of facilities available in medical institutions that face limited diagnostic facilities and depend on common laboratory assays for their diagnosis.

4.1. Model performance comparison

The supervised machine learning models are contrasted and

evaluated in Table 1. Overall, ensemble-based algorithms outperformed linear and kernel-based algorithms in terms of prediction capability. Among them, Extreme Gradient Boosting was seen to be the most effective and reliable algorithm that consistently produced consistently high results in the metrics evaluated being tested, highlighting the capability of handling complex nonlinear relationships between the different hematological features. The excellent functionality of Extreme Gradient Boosting may be explained by the fact that it is optimized by gradients, can model nonlinear interactions among features, and that it could withstand collinearity among correlated hematological data that are typical of clinical laboratory data. Also, its boosting mechanism gradually rectifies false-classified samples, allowing more accurate separation of overlapping classes of severity. In contrast, Logistic Regression and Support Vector Machine showed a poor performance, which highlights their shortcoming in modeling the overlap of features distribution inherent in multiclass clinical dataset, while Random Forest showed a good ability to predict as well.

These results highlight the suitability of Extreme Gradient Boosting algorithm for triage-level dengue severity prediction, particularly when relying solely on complete blood count parameters. Based on balanced performance across severity classes we will optimize the model selection, rather than solely on overall accuracy, which have been proved important clinically meaningful severity classification.

Although Random Forest achieved slightly higher weighted accuracy, Extreme Gradient Boosting demonstrated superior macro-level precision and F1-score, indicating better consistent performance across all severity classes, which is essential for clinically balanced triage decisions.

Table 2. Performance comparison of machine learning models.

Model	Accuracy (Macro)	Precision (Macro)	Recall (Macro)	F1-score (Macro)	F1-score (Weighted)
XGBoost	0.641148	0.592870	0.613878	0.591035	0.654796
Random Forest	0.655502	0.583768	0.585019	0.572324	0.658250
SVM	0.588517	0.584690	0.609303	0.560386	0.613342
Logistic Regression	0.583732	0.544700	0.551425	0.530919	0.608798
Page number	Publication team will format and assign page numbers				

Source: <https://github.com/Muqaddar45/DengueSeverityPrediction.git>

4.2. Triage-level severity classification analysis

The top-performing model was eventually evaluated at the class level to evaluate clinical relevance, as illustrated in Table 2 below. Strong recall for the severe dengue category was demonstrated using the model, which is clinically significant because missing to identify high-risk patients may result in delayed care and negative findings. The comparatively lower accuracy of the severe category can be attributed to the imbalance of the classes since the severe cases of dengue were a smaller fraction of the data, which predisposes the false-positive errors in this group. This is a tradeoff between precision and recall, which is clinically tolerable in the triage situation where it is

important that sensitivity consider severe cases to ensure that high-risk patients are not underestimated. Due to the inherent uncertain nature of intermediate disease progression, moderate patients almost overlapped with nearby severity levels. The model's ability to reduce unnecessary hospital admissions and optimize resource use has been confirmed by the more accurate classification of minor cases.

Misclassification patterns further suggest that moderate and severe cases share overlapping hematological profiles during early disease progression, emphasizing the inherent clinical complexity rather than model inadequacy.

Compared to binary severity classification techniques, the



implementation of a tri-level severity model provides a more comprehensive and clinically relevant assessment. These results indicate that, when investigated with a suitable machine learning technique, frequently accessible hematological features can help effective early risk classification.

Table 3. Class-wise performance of the best-performing model (XGBoost).

Severity Level	Precision	Recall	F1-score	Support
Minor	0.870588	0.643478	0.740000	115
Moderate	0.542169	0.714286	0.616438	63
Severe	0.365854	0.483871	0.416667	31

Source: <https://github.com/Muqaddar45/DengueSeverityPrediction.git>

The confusion matrix of XGBoost model for tri-level dengue severity classification is illustrated in Figure 1. Only a small percentage of severe cases are mistakenly labeled as moderate, and very few are mistakenly classified as minor. The model effectively recognizes severe dengue in addition to properly identifying the majority of mild and moderate cases. Because it reduces the possibility of underestimating high-risk patients and supports the model's appropriateness for triage-level clinical decision support, this behavior is clinically significant.

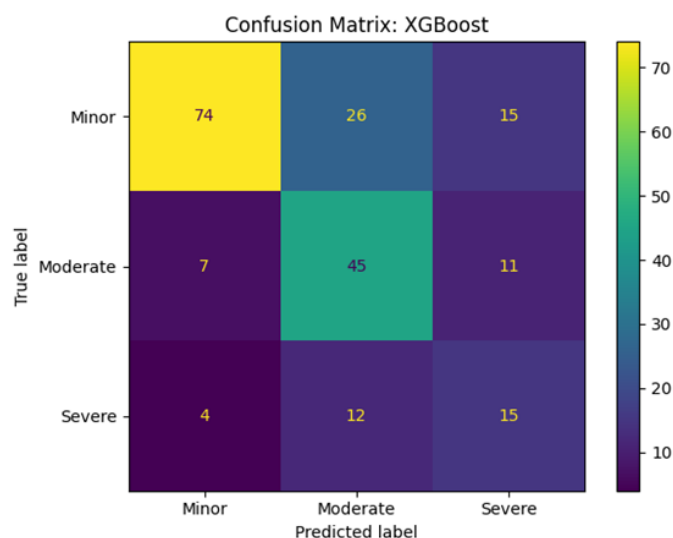


Figure 1. confusion matrix for tri-level for XGBoost Algorithm

4.3. Feature importance and clinical interpretability

To develop explainable ML model, these models were evaluating the clinical interpretability of the selected model. According to the SHAP-based feature importance analysis shown in Figure 1, the most significant predictors of dengue severity are platelet count, hematocrit, and white blood cell count. These features are well-known markers of dengue disease progression, with hemoconcentration and thrombocytopenia closely associated with plasma leakage and the development of severe illness. The interpretability framework facilitates the integration of the suggested model into a clinical decision support system and

increases clinical trust by providing a clear understanding of feature contributions. Because it enables doctors to understand and verify model predictions using well-established laboratory indicators, this interpretability is especially crucial for real-world adoption.

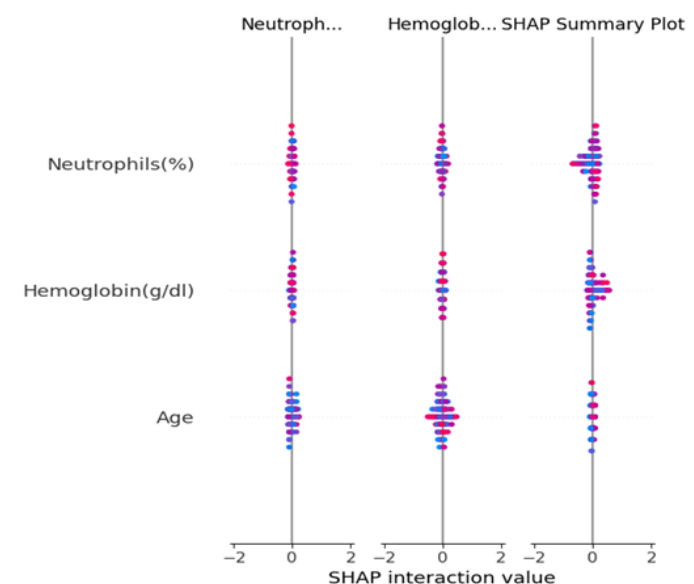


Figure 2. SHAP-based feature importance

4.4. Comparison with recent studies and clinical implications

The approach recommended has a number of advantages over recent dengue severity prediction research. Many existing models depend on binary classification systems, demographic elements, or serological features, which limits their use in earlier triage and resource-constrained settings. This study, on the other hand, presents a tri-level severity classification system that is more in line with actual clinical decision-making and exclusively uses regular complete blood count values. The results demonstrate that hematology-driven machine learning models can help medical professionals with resource allocation, triage prioritization, and early evaluation of dengue severity. The suggested approach has great promise for implementation in low-resource healthcare settings by leveraging readily accessible, reasonably priced laboratory tests.

4.5. Limitations and future directions

Although the research yielded the best conclusions, it is restricted to a specific publicly available dataset and to a retrospective examination. To further assess generalizability and clinical consequences, the forthcoming study should focus on an evaluation, integration into real-time clinical decision support systems, and external validation using multicenter datasets.

5. CONCLUSION

This study demonstrates how routinely available hematological data is effective in assisting triage-level dengue severity prediction when examined using machine learning algorithms.



The suggested algorithm effectively distinguishes between minor, moderate, and severe dengue cases by creating a hematology-driven, multi-class classification model. It also shows that an ensemble-based model is the most trustworthy model for this purpose. By elucidating the role of significant blood components, explainable artificial intelligence improves clinical interpretability, increasing trust and usefulness in clinical settings. The developments reveal the relevance of this model for clinical decision support and earlier threat assessment, especially for healthcare settings with limited resources and no access to state-of-the-art diagnostic tools. Despite its benefits, the study is denied by the lack of external verification and its reliance on a single available dataset. To additionally assess the model's impact on patient results and healthcare efficiency, forthcoming research should focus on confirming it across a range of clinical populations, adding longitudinal data, and ensemble the models into practical clinical decision-support systems.

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